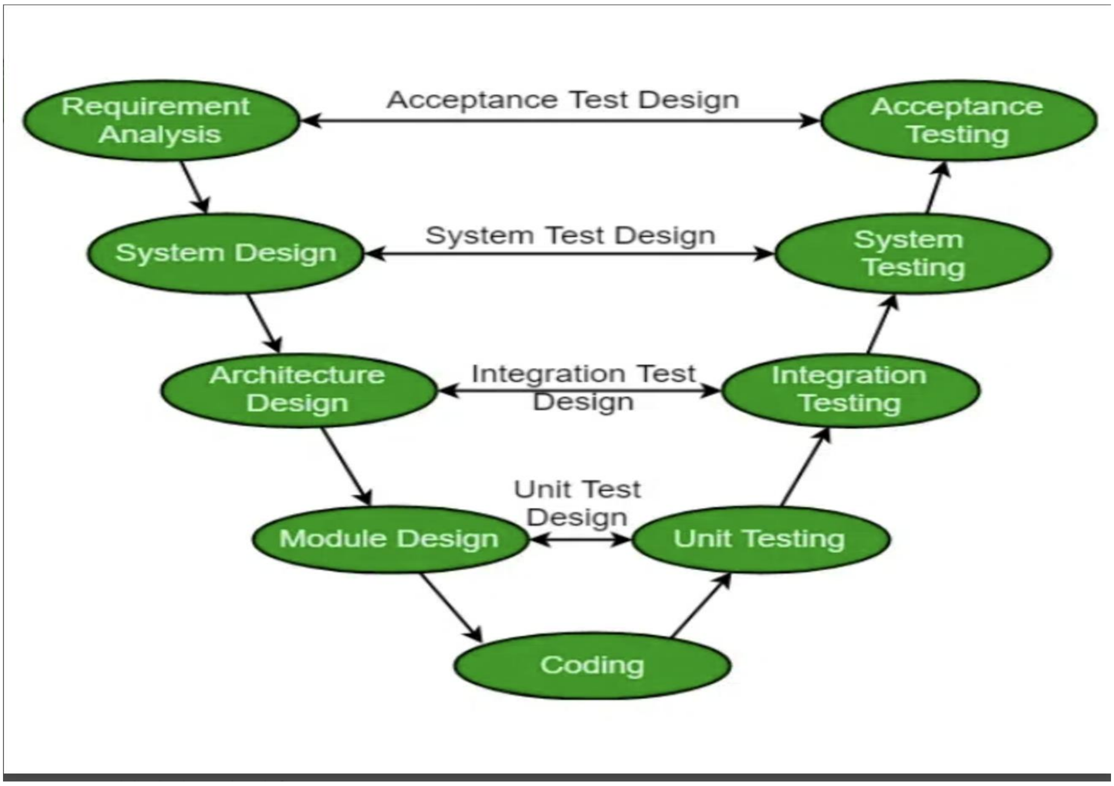
Assignment-2 Session3

Question 1: Draw the V-Model diagram on paper or using any digital tool, and clearly label each phase with both the developer and tester roles involved at each step.

Answer

V-Model (Verification and Validation Model)



Roles Involved

Development Phase	Developer Role	Corresponding Testing Phase	Tester Role
Business Requirements	Gather business requirements	Acceptance Testing	Verify business needs are satisfied
System Requirements	Prepare system specifications	System Testing	Verify complete system functionality
High Level Design	Design system architecture	Integration Testing	Verify interaction between modules
Low Level Design	Design individual modules	Unit Testing	Verify each module works correctly
Coding	Develop the application	Execute all testing phases	Identify and report defects

Question 2: Pick any app you use daily (like Zomato, Instagram, or Paytm) and write a short table mapping each SDLC phase to a real activity that would happen for that app.

Answer

Application Chosen: Zomato

SDLC Phase	Real Activity in Zomato
Requirement Gathering	Collect requirements for a new "Group Order" feature.
Design	Create UI screens and database design for group ordering.
Coding	Developers implement the feature using the approved design.
Testing	QA team verifies group creation, invitations, payments, and order placement.
Deployment	Release the feature to production through an app update.
Maintenance	Fix reported bugs, improve performance, and add enhancements based on user feedback.

Question 3: For each phase of the V-Model, list one specific testing activity that should be performed, using examples from a food delivery app.

Answer

V-Model Phase	Testing Activity	Example
Requirement Analysis	Review business requirements	Verify that users should be able to place food orders using multiple payment methods.
System Design	Prepare System Test Cases	Design test cases for complete order placement and delivery tracking.
High Level Design	Prepare Integration Test Cases	Verify communication between Cart, Payment, and Order modules.
Low Level Design	Prepare Unit Test Cases	Verify the discount calculation function returns the correct amount.
Coding	Developers complete implementation	Code is ready for testing.
Unit Testing	Test individual functions	Verify the promo code calculation module works correctly.
Integration Testing	Test module interaction	Verify Cart, Payment, and Order History work together.
System Testing	Test complete application	Verify a customer can search a restaurant, place an order, make payment, and track delivery successfully.
Acceptance Testing	Validate business requirements	Customer verifies the application meets all agreed business requirements before release.

Question 4: Compare the V-Model and RAD Model by listing 3 strengths and 2 weaknesses of each, focusing on how testing is handled in both models.

Answer

Feature	V-Model	RAD Model
Testing Approach	Testing is planned from the beginning and mapped to every development phase.	Testing is performed after each rapid prototype or iteration.
Development Speed	Moderate	Very Fast
Best Suited For	Banking, Healthcare, Government, and Safety-Critical Applications	Small to Medium Business Applications with frequently changing requirements

V-Model

Strengths

- Testing begins early in the software development life cycle.
- Every development phase has a corresponding testing phase, improving quality.
- Defects are identified earlier, reducing the cost of fixing them.

Weaknesses

- Difficult to accommodate changing requirements.
- Less suitable for projects where requirements frequently change.

RAD (Rapid Application Development) Model

Strengths

✧ Rapid development and faster product delivery.
✧ Continuous user feedback improves the final product.
✧ Changes can be implemented quickly during development.

Weaknesses

✧ Requires highly skilled developers and active customer involvement.
✧ Not suitable for very large or highly complex systems.